UCEST105 - ALGORITHMIC THINKING WITH PYTHON

Question Bank

Module 1 - Problem-Solving Strategies and Python Basics

Easy (11)

No.	Question	Marks
1	Work out the value of each Python expression: (i) `(50 - 5 * 6) // 4` (ii) `not (10 < 5) or (5 == 5)` (iii) `True and 5`	3
2	Evaluate the following and write the final result: (i) `(5 > 3) and (8 < 6) or (7 == 7)` (ii) `2**3**2` (iii) `15 and -8`	3
3	State the output of: (i) `(8 + 6 * 2 ** 3**1) / 7 - 4` (ii) `8 > 5 and 4 <= 4 or 25 != 30`	3
4	Give two examples of ill-defined problems and two examples of well-defined problems.	3
5	Imagine you must assemble a jigsaw puzzle without the picture of the finished image. Describe how you would complete it using trial and error.	3
6	For the input string "12345", write a short Python snippet that checks whether it is numeric.	3
7	Name any three functions of Python's math library and explain what each one does.	3
8	List four modules that come built into Python.	2
9	What are the basic data types in Python? Give an example of each.	3
10	The area of a circle is given. Write a Python program that uses the math module to find the radius.	3
11	Write a Python program that takes a time in minutes and displays it in hours and minutes (for example, 85 minutes is 1 hour 25 minutes).	4

Medium (11)

No.	Question	Marks
12	Write a Python program, without using any loops, to reverse a three-digit number.	3
13	How do algorithms differ from heuristics in problem solving? Give an example in which a heuristic reaches a solution faster than an algorithm.	5

No.	Question	Marks
14	Choose the most suitable problem-solving strategy for each case and justify your choice: (i) You are stuck near the last few empty cells of a difficult Sudoku puzzle. (ii) You must plan a family trip (pick a destination, book flights, find accommodation) within a fixed budget.	3
15	Explain how a heuristic strategy can help you prepare a time-management plan for studying several subjects for your second-semester university examinations.	3
16	You are new to a large city and want to go out for dinner with friends. Explain how a heuristic approach can help you find the best restaurant.	4
17	Describe how the backtracking strategy can be used to solve a Sudoku puzzle.	5
18	You have been asked to develop new campus management software for your college. Explain how you would apply Means-Ends Analysis.	6
19	As a fitness expert you want to write and publish a book of customised workout routines that suits different fitness levels and is easy to follow, all within one year. Show how Means-Ends Analysis can help you reach this goal.	7
20	For each of the approaches (1) trial and error, (2) heuristic approach and (3) means-ends analysis, give two real-world situations where it can be applied effectively, and justify each.	6
21	Walk through the six steps of the problem-solving process using the task of finding the area of a circle.	6
22	Apply the six problem-solving steps to the task of finding the area of a right-angled triangle.	9

Hard (1)

No.	Question	Marks
23	Apply the six steps of the problem-solving process to the task of finding the real roots of a quadratic equation.	6

Module 2 - Algorithms, Pseudocode and Flowcharts

Easy (7)

No.	Question	Marks
24	Write pseudocode to find the sum of all even numbers from 1 to N.	3
25	Draw a flowchart that calculates the factorial of a number.	3
26	Draw a flowchart that prints all the integers below 50.	3
27	Write an algorithm that displays all odd numbers between 1 and 500 in descending order.	3
28	Define pseudocode. State any two advantages of using pseudocode while designing algorithms instead of writing the code directly.	3

No.	Question	Marks
29	Write pseudocode that finds the factorial of a given number using a for loop.	3
30	A logistics company keeps a list of delivery times in minutes. Write an algorithm that prints those times that are divisible by 6 but not divisible by 4.	3

Medium (13)

No.	Question	Marks
31	Draw a flowchart to generate the first n numbers of the Fibonacci sequence.	3
32	Write an algorithm to test whether an input integer is a palindrome.	3
33	Draw a flowchart for a parking-lot fee calculator. It should ask for the vehicle type (car, bike or truck) and the parking duration in hours, apply an hourly rate that you decide for each vehicle type, and output the final fee.	3
34	A business analytics tool must find the highest and lowest values in a list of N sales figures. Write efficient pseudocode to find the largest and smallest numbers.	6
35	Write pseudocode to compute the average height of the boys and the average height of the girls in a class of n students.	6
36	Draw a flowchart that uses a CASE construct to display the day of the week for a number from 1 to 7.	4
37	Draw a flowchart, using a for loop, to find the largest of n numbers that are entered one after another.	5
38	Write pseudocode that keeps reading numbers until the value -5555 is entered, and then displays the average of only those numbers that are divisible by 4 (excluding -5555). If no number divisible by 4 was entered, display a suitable message.	5
39	A farmer divides a farm into six sections and records the quantity of crop harvested and the area harvested for each section. Draw a flowchart to compute the average yield per hectare for the whole farm.	4
40	Draw a flowchart for a traffic-light control system. The input is a colour (Red, Yellow or Green) and the output is the action ("Stop", "Prepare to stop" or "Go"). Show an error message for any other colour.	4
41	During the festive season a shop gives a 10% discount on all mobile phones and an extra flat exchange discount of 1000 for an old phone (not every customer has one). Draw a flowchart that reads the original price and prints the selling price.	5
42	Given the three sides of a triangle, write an algorithm or pseudocode to decide whether it is isosceles, equilateral or scalene.	4
43	Write pseudocode to find the average age of the students in a class. The user ends the input by entering 0.	5

Hard (2)

No.	Question	Marks
44	Write pseudocode for a grading system with these rules: marks of 90 and above get A; 80-89 get B; 70-79 get C; 60-69 get D; 50-59 get E; below 50 get F. If the difference between the marks and the next multiple of 10 is less than 3, raise the grade to the next higher grade. No such raising is done for marks below 50.	6
45	A robot moves on an N x N grid whose top-left cell is (0,0). It can move up, down, left or right but must never leave the grid. Write an algorithm that takes a starting position and a list of commands ("up", "down", "left", "right") and finds the robot's final position. Example: on a 5x5 grid, starting at (2,2) with commands ["left", "down", "right"], the robot ends at (3,2).	5

Module 3 - Python Programming: Control Flow, Data Structures, Functions and Recursion

Easy (9)

No.	Question	Marks
46	Write the output of each of the following: (i) `for count in range(-10, -20, -2): print(count, end=" ")` (ii) `for i in range(10, -1, -2): print(i)` (iii) `2*3**2**2`	3
47	Write a Python program to find the sum of the even numbers among N given numbers.	3
48	Let `s = "MALAYALAM"`. Write the output of the following slicing operations: (i) `s[-3:]` (ii) `s[::-1]` (iii) `s[1:8:2]`	3
49	Given `numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90]`, write slicing expressions to: (1) take the first 5 elements, (2) take every second element starting from the first, (3) reverse the list.	3
50	What is modularization in software development? Discuss its benefits.	3
51	Explain, with a suitable example, any one way of returning multiple values from a Python function.	3
52	Write a Python program that reads n numbers into a list and displays the sum of the cubes of its elements.	3
53	Compare a set and a list in Python on any two features. Write code to convert `my_list = [1, 2, 2, 3, 4, 4, 5]` into a set, and state what you observe when the set is printed.	4
54	What are the main differences between lists, tuples and strings in Python with respect to mutability, usage and methods?	3

Medium (11)

No.	Question	Marks
55	What is circularity in recursion? How can it be prevented in recursive functions?	3
56	Write a recursive function that searches for a key in a list of N integers. It is first called with the list, the key and the starting index as arguments, and it returns the index of the key if found, otherwise -1.	3
57	A financial analysis company models growth patterns using the Fibonacci sequence. Write a recursive solution that generates the first n Fibonacci numbers.	5
58	Write a Python program that accepts the lengths of three sides of a triangle and determines whether it is a right-angled triangle.	4
59	Write a Python program to create a one-dimensional NumPy array and reverse its contents without using a second array.	3
60	To register for the end-semester examination you must log in to the university portal. Write a Python program to validate the credentials, assuming usernames are stored in a list USERNAME and the matching passwords in a list PASSWORD, so that PASSWORD[i] belongs to USERNAME[i].	5
61	Write a recursive Python function to add two positive numbers.	4

62	Study the recursive function below, which takes a non-negative integer.
	(1) Describe what the function computes.
	(2) Explain how the call stack grows and shrinks during the computation for the input 123.
	`def function_1(n):`
	`  if n == 0:`
	`  return 0`
	`  else:`
	`  num = n % 10`
	`  smallerNumber = n // 10`
	`  return num + function_1(smallerNumber)`
	`print(function_1(123))`
		5

63	Write a Python function `isValid_mobile_no` that checks whether a number is a valid mobile number. The number must contain exactly 10 digits and its first digit must be 7, 8 or 9.	3
64	Write a recursive Python function to find the GCD of two numbers.	3
65	Write a recursive function to multiply two positive numbers. Trace the function calls for multiplying 3 and 4 and explain how the call stack changes during execution.	5

Hard (5)

No.	Question	Marks
66	Given two positive integers a and b, write a recursive Python program to find their LCM. Do not apply a direct LCM formula; instead use the relationship lcm(a, b) x gcd(a, b) = a x b together with a recursive GCD.	6

No.	Question	Marks
67	A content-moderation tool must rearrange the characters of a sentence so that capital letters come first, then small letters, then digits, and finally special characters. Write a Python program that displays the rearranged output as a single word.	3
68	Write a menu-driven Python program that maintains a telephone directory using a dictionary. The operations are: add contact, update contact, delete contact, display all contacts and exit.	6
69	A twisted prime is a prime number whose reverse is also a prime number. Write a Python program that checks whether a given number is a twisted prime.	6
70	Write a Python program that reads two lists from the user and merges them into a third list in which all even numbers come first, followed by all odd numbers, with both the even and the odd numbers in sorted order.	4

Module 4 - Algorithmic Approaches: Brute Force, Divide and Conquer, Greedy, Dynamic Programming and Randomized Methods

Easy (5)

No.	Question	Marks
71	What is the motivation for using a randomized approach in problem solving?	3
72	Compare Divide and Conquer with Dynamic Programming.	3
73	List any three advantages and any three disadvantages of the divide and conquer approach.	3
74	What do memoization and tabulation mean in problem solving, and how do they differ? (No programs needed.)	3
75	What is the brute-force approach to computational problem solving? What is the maximum number of attempts needed to guarantee opening a four-digit numeric padlock (digits 0 to 9) using it?	3

Medium (13)

No.	Question	Marks
76	A security application uses a 4-digit numeric password, and a user has forgotten theirs because of a system error. Write an algorithm to recover the password.	3
77	Compare greedy algorithms and dynamic programming in terms of approach, guarantee of an optimal solution and suitable scenarios. Give examples where each is the preferred choice.	4
78	Explain the memoization and tabulation techniques of dynamic programming by using them to calculate the n-th Fibonacci number.	6
79	You are given an array of positive integers in which each value is the time needed for a task. Develop a greedy algorithm to find the maximum number of tasks that can be completed within a given total time limit, explaining the reasoning behind each step.	3

No.	Question	Marks
80	For what type of problems is the greedy method guaranteed to give a correct (optimal) answer? Explain with a suitable example.	3
81	The Subset Sum problem asks whether some subset of a given set of numbers adds up to a target sum. For example, in {2, 4, 6, 8} with target 10, the subset {4, 6} works. Write a brute-force approach to solve it.	5
82	Write the bottom-up dynamic programming algorithm to compute the n-th Fibonacci number.	4
83	Explain and illustrate the Merge Sort procedure by applying it to the list 38, 27, 43, 3, 9, 82, 10.	6
84	Identify a real-world scenario in which brute force is preferred over divide and conquer, and explain why brute force suits it better.	3
85	Write a greedy algorithm for this problem: given an array of positive integers representing item weights and a bag with a maximum weight limit, maximise the number of items placed in the bag without exceeding the limit. Example: items = [3, 2, 5, 1, 4, 6], max_weight = 10 gives a maximum of 4 items.	6
86	Compare Dynamic Programming and Recursion in terms of approach, efficiency and use cases.	3
87	Illustrate how the array [15, 8, 3, 12, 6, 10, 4, 1] is sorted using merge sort. Draw a diagram showing how the array is split and merged at each stage.	6
88	A vending machine has an infinite supply of coins of given denominations and must return change using the fewest coins. Implement a greedy algorithm that finds the coin combination with the minimum number of coins for a given amount.	5

Hard (3)

No.	Question	Marks
89	A bank has a list of loan interest rates (positive integers) from several branches. Using the Divide and Conquer approach, develop an efficient algorithm to find the sum of the two smallest rates, and explain it with a suitable example.	6
90	Explain how a randomized method can be used to approximate the expected number of people who get their own hat back when n hats are handed out at random among n people.	3
91	Ram wants to count the distinct ways to climb a staircase of N steps, taking 1 or 2 steps at a time (for N = 3 the ways are (1,1,1), (1,2) and (2,1), so the answer is 3). Identify the optimal substructure and the overlapping subproblems in this problem to show that dynamic programming can be applied.	6

Part B: 100 New Practice Questions

Module 1 - Problem-Solving Strategies and Python Basics

Easy (9)

No.	Question	Marks
1	Define a problem. Differentiate between a well-defined and an ill-defined problem with one example each.	3
2	List, in order, the six steps of the problem-solving process.	3
3	Define an algorithm and state any three characteristics of a good algorithm.	3
4	What is the trial and error method? Give two everyday situations where it is used.	3
5	Evaluate the following Python expressions: (i) `17 // 5 + 17 % 5` (ii) `2 ** 2 ** 3` (iii) `not (4 > 2 and 3 < 1)`	3
6	Write Python statements to read two integers from the user and print their sum and product.	3
7	What is type conversion in Python? Write the result of `int("45") + 3`, `str(45) + "3"` and `float(7) / 2`.	3
8	What is a heuristic? Give one example of a rule of thumb that people use in daily life.	3
9	Write a Python program that converts a temperature given in Celsius to Fahrenheit.	3

Medium (9)

No.	Question	Marks
10	Explain the working-backwards strategy with a suitable example.	4
11	Differentiate between an algorithm and a heuristic by comparing two ways of finding a particular book in a large library.	5
12	A student wants to reach college on time every day and has three possible routes. Apply the six steps of problem solving to decide the best route.	6
13	Explain how Means-Ends Analysis can be used to plan and organise a three-day technical fest in your college.	6
14	Compare trial and error with backtracking. Which is more systematic, and why?	4
15	Write a Python program to read the principal, rate and time, and compute the simple interest and the compound interest.	4
16	Write a Python program that reads the coordinates of two points (x1, y1) and (x2, y2) and finds the distance between them using the math module.	4
17	Explain, with an example such as organising an inter-college sports meet, how decomposing a complex problem into sub-problems helps in solving it.	5
18	Write a Python program to find the sum of the digits of a three-digit number without using loops.	4

Hard (4)

No.	Question	Marks
19	Explain how backtracking can be used to solve the 4-Queens problem. Show at least one dead end and the backtracking step that follows it.	6
20	Using means-ends analysis (or a state-space approach), show how to measure exactly 4 litres of water using only a 5-litre jug and a 3-litre jug. List each step and the state after it.	6
21	A cab-booking app must choose a driver for each ride request. Discuss how a heuristic can give a good (but not guaranteed best) choice, and describe a situation where your heuristic would fail.	6
22	Write a Python program that reads the three sides of a triangle, checks that they can form a triangle, and then finds the area using Heron's formula and the math module.	6

Module 2 - Algorithms, Pseudocode and Flowcharts

Easy (9)

No.	Question	Marks
23	Differentiate between an algorithm, a flowchart and pseudocode.	3
24	Name the commonly used flowchart symbols (terminator, process, decision, input/output, connector, flow line) and state the purpose of each.	3
25	Write pseudocode to read a number and print whether it is even or odd.	3
26	Draw a flowchart to find the sum of the first N natural numbers.	3
27	Write an algorithm to find the area of a rectangle given its length and breadth.	3
28	Write pseudocode to find the largest of three numbers using if-else statements.	3
29	Draw a flowchart to check whether a person is eligible to vote (age 18 or above).	3
30	Differentiate between the sequence, selection and iteration constructs with one example of each.	3
31	Write pseudocode to print the multiplication table of a given number up to 10.	3

Medium (11)

No.	Question	Marks
32	Draw a flowchart to count the number of digits in a positive integer.	4
33	Write pseudocode to find the sum of the digits of a given number.	4
34	Write an algorithm to check whether a given number is prime.	5
35	Draw a flowchart to print all the numbers between 1 and 100 that are divisible by 7.	4
36	Write pseudocode for an electricity bill calculator with slab rates: the first 100 units at Rs. 3 per unit, the next 100 units at Rs. 5 per unit, and any units above 200 at Rs. 7 per unit.	6
37	Write pseudocode that reads the marks of n students, computes the class average and counts how many students passed (marks of 40 or above).	5
38	Write an algorithm to reverse a given integer and display the result.	4
39	Draw a flowchart for a simple ATM PIN check that allows the user at most three attempts.	5

No.	Question	Marks
40	Write pseudocode to find the second largest number in a list of N numbers.	5
41	Draw a flowchart that uses a CASE construct to simulate a calculator performing +, -, * and / on two numbers, and shows an error message for division by zero.	5
42	Write pseudocode to count the number of vowels in a given sentence.	4

Hard (6)

No.	Question	Marks
43	Write pseudocode to check whether a given number is an Armstrong number (it may have any number of digits).	6
44	Write an algorithm to convert a decimal number into its binary equivalent, and trace it for the number 37.	6
45	Write pseudocode to print all the leap years between two given years.	6
46	Draw a flowchart for Euclid's algorithm to find the GCD of two numbers and trace it for 48 and 18.	6
47	Write pseudocode for bubble sort on n numbers and show the first pass for the list 5, 1, 4, 2, 8.	7
48	Design an algorithm and flowchart for a library fine system: Rs. 2 per day for the first 7 days of delay, Rs. 5 per day after that, and the fine can never exceed Rs. 500.	7

Module 3 - Python Programming: Control Flow, Data Structures, Functions and Recursion

Easy (11)

No.	Question	Marks
49	Predict the output of: (i) `for i in range(1, 10, 3): print(i, end=" ")` (ii) `for j in range(5, 0, -2): print(j)`	3
50	Let `s = "Programming"`. Write the output of `s[2:6]`, `s[::2]`, `s[-4:]` and `len(s)`.	3
51	Differentiate between a list and a tuple in Python (any three differences).	3
52	Write a Python program to count the number of vowels in a string.	3
53	Given `d = {"a": 1, "b": 2}`, write statements to add the key "c" with value 3, change the value of "a" to 10, delete the key "b" and print all the keys.	3
54	Write Python statements that find the sum, maximum, minimum and length of a list of numbers using built-in functions.	3
55	What is a function? Write a function that returns the square of a number and show how to call it.	3
56	Write a Python program to print the multiplication table of a number using a for loop.	3
57	Differentiate between a set and a dictionary in Python.	3
58	What is a base case in recursion? Why is it essential?	3
59	Write a Python program to check whether a number is positive, negative or zero using if-elif-else.	3

Medium (12)

No.	Question	Marks
60	Write a Python program to check whether a string is a palindrome, ignoring the case of letters.	4
61	Write a Python program that reads n numbers into a list and prints the second largest number without using sort().	5
62	Write a Python program to count the frequency of each word in a sentence using a dictionary.	5
63	Write a recursive function to find the factorial of n and trace the call stack for n = 4.	5
64	Write a recursive function to compute x raised to the power n (n is a non-negative integer).	4
65	Write a recursive function to reverse a string.	4
66	Write a function that returns both the minimum and the maximum of a list, and show how the returned values are unpacked by the caller.	4
67	Create a 3 x 3 NumPy array containing the numbers 1 to 9. Write statements to print the sum of each row, the sum of each column and the sum of the main diagonal.	5
68	Given two lists, write Python statements to find their common elements, their union and the elements present only in the first list, using sets.	4
69	Write a Python program to remove duplicate elements from a list while keeping the original order.	4
70	Differentiate between local and global variables with a program, and show its output.	4
71	Write a Python program to count the uppercase letters, lowercase letters, digits and special characters in a string.	5

Hard (7)

No.	Question	Marks
72	Write a recursive function to find the n-th Fibonacci number. Draw the recursion tree for fib(5), count the number of function calls and explain why this approach is inefficient.	7
73	Write a recursive binary search function and trace it while searching for 23 in the list [2, 5, 8, 12, 16, 23, 38, 56].	7
74	Write a menu-driven program that stores student records in a dictionary (name mapped to a list of three marks). It should add a student, display the topper and display the class average.	7
75	Write a Python program to check whether two strings are anagrams without using sort().	6
76	Write a recursive function to solve the Tower of Hanoi puzzle for n disks and trace its output for n = 3.	7
77	Write a Python function that merges two sorted lists into a single sorted list without using sort().	6
78	Write a Python program to read an m x n matrix (list of lists) and print its transpose.	6

Module 4 - Algorithmic Approaches: Brute Force, Divide and Conquer, Greedy, Dynamic Programming and Randomized Methods

Easy (8)

No.	Question	Marks
79	State the main idea of divide and conquer and list its three steps.	3
80	What is a brute-force approach? Give one advantage and one disadvantage.	3
81	Define optimal substructure and overlapping subproblems with a short example.	3
82	Explain the greedy-choice idea with a real-life example.	3
83	What is a randomized algorithm? Give one example where randomness helps.	3
84	Compare the top-down and bottom-up approaches in dynamic programming.	3
85	How many comparisons does linear search need in the worst case on 1024 elements? How many does binary search need on the same sorted list, and why?	3
86	Write a brute-force Python snippet to find all pairs of elements in a list whose sum equals a given target.	3

Medium (9)

No.	Question	Marks
87	Explain the working of merge sort and state its time complexity. Show the merge step for the sorted lists [3, 8] and [1, 9].	5
88	Explain how the maximum element of an array can be found using divide and conquer, and apply it to [4, 9, 2, 7, 5, 1].	5
89	Use a greedy strategy to select the maximum number of non-overlapping activities from (start, finish): (1,3), (2,5), (4,7), (1,8), (5,9), (8,10). Justify your selection rule.	6
90	Use the greedy method to make change for 63 with coins {1, 5, 10, 25}. Then show, with coins {1, 3, 4} and amount 6, that the greedy method can fail.	6
91	Solve the fractional knapsack problem greedily for items (value, weight) = (60, 10), (100, 20), (120, 30) with capacity 50.	6
92	Write a Python function that computes the n-th Fibonacci number using memoization with a dictionary, and explain how it avoids repeated computation.	5
93	Write a Python program that uses tabulation to count the ways of climbing N stairs when 1, 2 or 3 steps can be taken at a time.	5
94	Explain how a randomized (Monte Carlo) method can estimate the value of pi using random points in a unit square, and outline a Python program for it.	5
95	Explain brute-force pattern matching and trace it for searching the pattern "ABC" in the text "ABABCABC".	5

Hard (5)

No.	Question	Marks
96	Write a Python function for quick sort using divide and conquer and trace it on the list [10, 7, 8, 9, 1, 5].	7
97	Solve the 0/1 knapsack problem using dynamic programming for weights {1, 3, 4, 5}, values {1, 4, 5, 7} and capacity 7. Show the table that is filled.	8

No.	Question	Marks
98	For coin denominations {1, 3, 4} and amount 6, compare the greedy, brute-force and dynamic programming approaches. Write a tabulation-based Python program that finds the minimum number of coins.	8
99	Find the Longest Common Subsequence of "ABCBDAB" and "BDCABA" using dynamic programming and show how the table is built.	8
100	Write a simulation program that estimates the probability that at least two people in a group of 23 share a birthday. Explain how the number of trials affects the accuracy of the estimate.	7